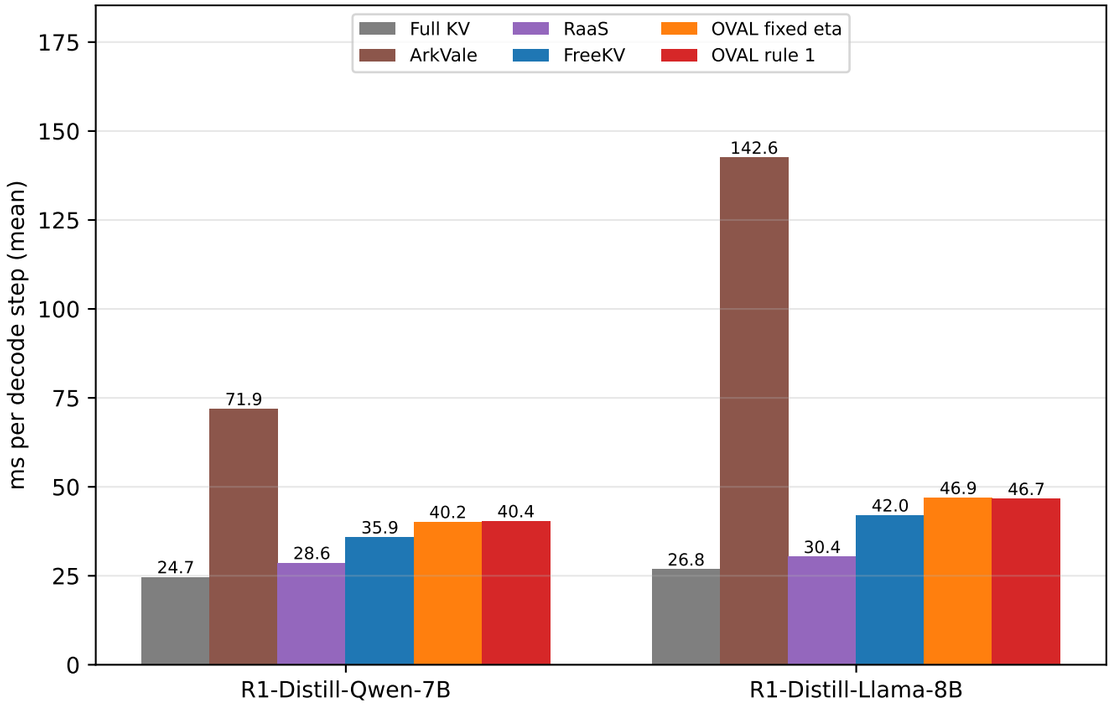
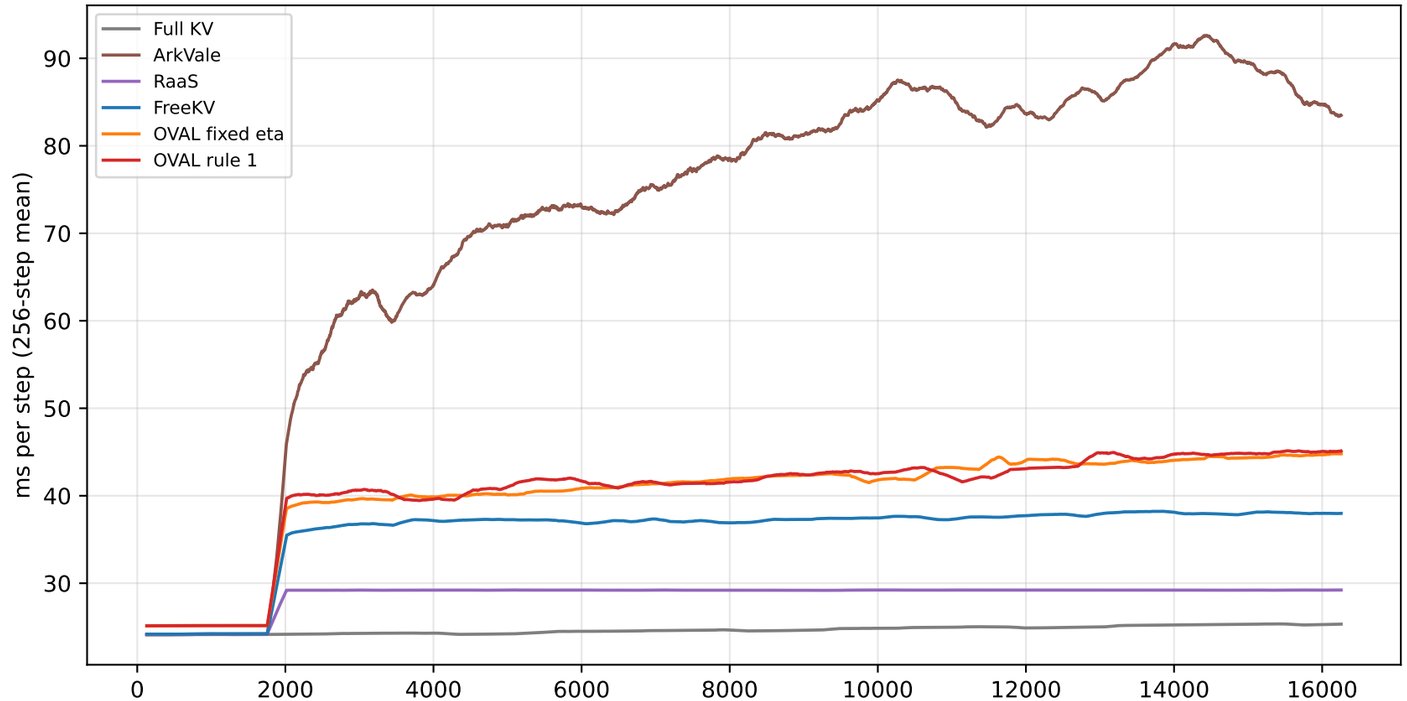
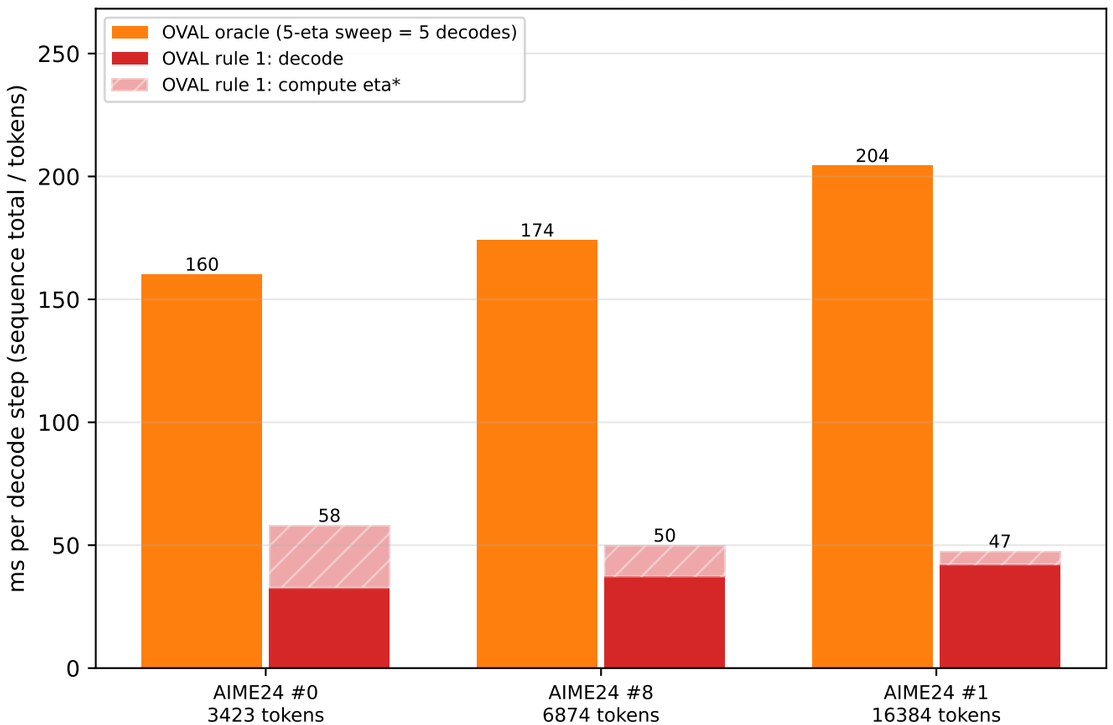
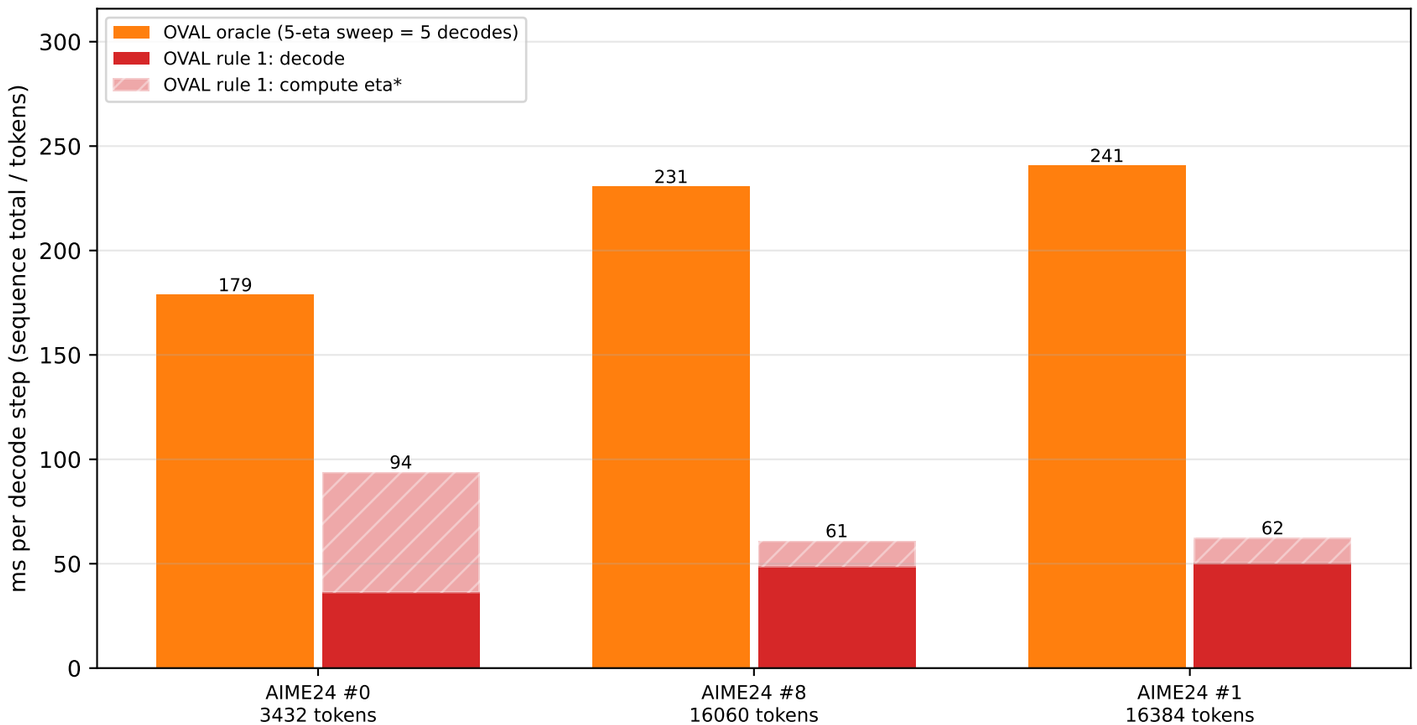

Decode latency, AIME24 problem 0, 16384 decoded tokens, batch 1
budget 2048 tokens (incl. 512 sink + 512 recent), A6000
Full KV
ArkVale
RaaS
FreeKV
OVAL fixed eta
OVAL rule 1
ms per decode step (mean)
0
25
50
75
100
125
150
175
24.7
71.9
28.6
35.9
40.2
40.4
26.8
142.6
30.4
42.0
46.9
46.7
R1-Distill-Qwen-7B
R1-Distill-Llama-8B


R1-Distill-Qwen-7B: per-step decode time over the generation
Full KV
ArkVale
RaaS
FreeKV
OVAL fixed eta
OVAL rule 1
ms per step (256-step mean)
30
40
50
60
70
80
90
0
2000
4000
6000
8000
10000
12000
14000
16000
decode step


R1-Distill-Qwen-7B: OVAL oracle (sweep 5 etas) vs OVAL rule 1
OVAL oracle (5-eta sweep = 5 decodes)
OVAL rule 1: decode
OVAL rule 1: compute eta*
ms per decode step (sequence total / tokens)
0
50
100
150
200
250
160
58
174
50
204
47
AIME24 #0
3423 tokens
AIME24 #8
6874 tokens
AIME24 #1
16384 tokens


R1-Distill-Llama-8B: OVAL oracle (sweep 5 etas) vs OVAL rule 1
OVAL oracle (5-eta sweep = 5 decodes)
OVAL rule 1: decode
OVAL rule 1: compute eta*
ms per decode step (sequence total / tokens)
0
50
100
150
200
250
300
179
94
231
61
241
62
AIME24 #0
3432 tokens
AIME24 #8
16060 tokens
AIME24 #1
16384 tokens